

- Q.25 Describe the concept and importance of transition curves in road construction. (CO6)
- Q.26 How is the earthwork for a reservoir calculated using contour maps? (CO2)
- Q.27 What are the different methods of traversing in theodolite surveying? (CO4)
- Q.28 Describe the temporary adjustments required for a transit theodolite. (CO4)
- Q.29 Explain the process of setting out a vertical curve and its application in road alignment. (CO6)
- Q.30 How is the height of an object with an inaccessible base measured using theodolite? (CO5)
- Q.31 What are the errors in theodolite surveying and the precautions taken to minimize them? (CO5)
- Q.32 Describe the working and applications of an EDM instrument in surveying. (CO9)
- Q.33 Discuss the procedure for making the alignment of a road on a contour map. (CO2)
- Q.34 Explain the concept of tachometry and its importance in hilly terrains. (CO3)
- Q.35 What is the use of Total station in modern surveying, and what are its advantages? (CO8)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain in detail the methods of contouring and the role of contour maps in civil engineering projects. (CO1)
- Q.37 Discuss the methods of setting out simple circular curves, Explain in detail any one. (CO6)
- Q.38 Describe in detail the use of modern surveying instruments such as EDM, Total Station, and GPS in surveying operations. (CO9)

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4th Sem / Civil, Constr, Mgmt., Highway Engg. Subject:- Surveying - II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the horizontal equivalent in contouring? (CO1)
- Difference in elevation between two contour lines
 - Horizontal distance between two contour lines
 - Inclination between contour lines
 - Vertical distance between two points
- Q.2 A line joining points of equal elevation on the ground is called a: (CO1)
- Level line
 - Contour line
 - Traverse line
 - Horizontal line
- Q.3 Theodolite is primarily used for measuring: (CO2)
- Vertical angles
 - Horizontal distances
 - Both horizontal and vertical angles
 - Horizontal angles only
- Q.4 The process of prolonging a line forward in theodolite surveying involves: (CO2)
- Taking a back sight reading
 - Measuring a horizontal angle

- c) Taking a fore sight reading
d) Shifting the instrument along the line
- Q.5 The stadia method of tachometry is based on the principle of: (CO3)
a) Triangulation b) Parallax
c) Optical wedge d) Similar triangles
- Q.6 A simple circular curve is generally set out using: (CO4)
a) Contour lines b) Vertical curves
c) Tangential angles d) Stadia readings
- Q.7 A transition curve is introduced between the straight and circular portions of a road to: (CO5)
a) Increase speed
b) Avoid accidents
c) Provide gradual change in curvature
d) Measure distances
- Q.8 Which instrument is used for the precise measurement of both horizontal and vertical distances in modern surveying? (CO6)
a) Auto level b) Theodolite
c) EDM d) Total station
- Q.9 In a vertical curve, the point of intersection of the tangents is known as : (CO8)
a) Tangent point b) Apex point
c) Transition point d) Point of curvature
- Q.10 The digital equipment used for plotting areas directly on maps is: (CO9)
a) Planimeter b) EDM
c) Total station d) Distomat

(2) 180744/170744/120744
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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The interval between successive contour lines on a map is called the _____. (CO1)
- Q.12 The process of measuring the height of an object from an inaccessible base is done using a _____. (CO2)
- Q.13 _____ curves are introduced to connect a straight and a curved path smoothly. (CO3)
- Q.14 In tachometric surveying, distances are calculated using the _____ method. (CO4)
- Q.15 A theodolite is temporarily adjusted by first leveling the _____. (CO5)
- Q.16 The point of tangency is also called the _____ point in curve setting. (CO6)
- Q.17 The _____ method is used for contouring in areas where direct observation is difficult. (CO6)
- Q.18 The vertical distance measured between two points on a map is called the _____ difference. (CO7)
- Q.19 A _____ is used to measure the angles and bearings in traverse surveying. (CO8)
- Q.20 The digital instrument used to determine areas on a contour map is known as a _____. (CO9)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the characteristics and importance of contours in civil engineering projects. (CO1)
- Q.22 Describe the method of interpolation of contours in direct contouring. (CO1)
- Q.23 What are the key elements of a simple circular curve, and how is it set out using theodolite? (CO2)
- Q.24 Explain the principle of tachometry and how it is used in modern surveying. (CO3)

(3) 180744/170744/120744
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